Type MS · Explain sex-linked inheritance and why males are affected more

Answer 1 of 7 · Question 7 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

7	(a) (i)	• when both present; • both (equally) expressed;		2
	(ii)	• I^BI^O; • I^BI^B;		2
	(b)	any four from • carried on X chromosome; • not expressed in heterozygotes/recessive; • can't have male carriers; • can only have female heterozygotes/carriers; • because only females can have 2X chromosomes • male XY/only one X chromosome;		4

Total 8 marks

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 1 of 7 · Question 7 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 2 of 7 · Question 6 part a.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number					
6	(a)	(i)	any three from • prevents excessive loss of blood (1) • forms scab over wound (1) • prevents entry of bacteria /pathogens/microorganisms(1) • so prevents infection (1) • allows wound to heal (1)		3
		(ii)	• (allele) on X/sex chromosome (1) • passed on to offspring/inherited by children (1)		2
	(b)	(i)	D; (X^hY) A the genotype is not X^hX^h B the genotype is not X^HX^h C the genotype is not X^HY		1
		(ii)	B; (X^HX^h) A the genotype is not $X^h X^h$ C the genotype is not X^HY D the genotype is not X^hY		1
		(iii)	• offspring genotypes X^HX^h and X^hY (1) • 1 carrier female to 1 haemophiliac male = 50%/0.5/½ (1)	full marks for correct figures only.	2
		(iv)	• unusual for female to be haemophiliac (1) • both alleles have to be affected (1) • must have a haemophiliac father and mother with haemophilia/carrier (1)	any correct reference to excessive blood loss	3

Total for question = 12 marks

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 2 of 7 · Question 6 part a.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 3 of 7 · Question 6 part b.iv · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.iv in this mark scheme extract

Number				
6 (a)	(i)	any three from • prevents excessive loss of blood (1) • forms scab over wound (1) • prevents entry of bacteria /pathogens/microorganisms(1) • so prevents infection (1) • allows wound to heal (1)		3
	(ii)	• (allele) on X/sex chromosome (1) • passed on to offspring/inherited by children (1)		2
(b)	(i)	D; (X^hY) A the genotype is not X^hX^h B the genotype is not X^HX^h C the genotype is not X^HY		1
	(ii)	B; (X^HX^h) A the genotype is not $X^h X^h$ C the genotype is not X^HY D the genotype is not X^hY		1
	(iii)	• offspring genotypes X^HX^h and X^hY (1) • 1 carrier female to 1 haemophiliac male = 50%/0.5/½ (1)	full marks for correct figures only.	2
	(iv)	• unusual for female to be haemophiliac (1) • both alleles have to be affected (1) • must have a haemophiliac father and mother with haemophilia/carrier (1)	any correct reference to excessive blood loss	3

Total for question = 12 marks

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 3 of 7 · Question 6 part b.iv · 3 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 4 of 7 · Question 8 part c · 5 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

8 (a)	any three from • change in base sequence/named mutation (1) • of DNA (1) • causing change in genotype/gene (1) • incorrect protein produced (1) • change in phenotype/characteristics (1)	Allow change in DNA for mps 1 and 2 (1)	3
(b) (i)	father X^HY mother X^HX^h (1) gametes X^H Y X^H X^h (1) F1 X^HX^H normal female; X^HX^h carrier female; X^HY normal male; X^hY haemophiliac (1) male		4
(ii)	(chance of first and second child is) ¼/0.25/25% (1) for both is ¼ x ¼/0.25 x 0.25 (1) = 1/16/0.0625/6.25% (1)		3
(c)	any five from • males only have one X chromosome (1) • two X chromosomes/H and h needed to be a carrier (1) • recessive allele/h (1) • carried on X chromosome (1) • either present or not (1) • no chance for dominant allele to be present (1) • to mask recessive (1)		5

Total 15 marks

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 4 of 7 · Question 8 part c · 5 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

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Type MS · Explain sex-linked inheritance and why males are affected more

Answer 5 of 7 · Question 9 part b.i · 2 marks

2024 JUN · Paper 01 · find part b.i in this mark scheme extract

9 (a)	(i)	A, B, AB;	2 marks for all 3 correct, 1 mark for 1 or 2 correct	(2)
	(ii)	any five from • blood type O; • no antigens present; • (person) contains anti-A and anti-B antibodies; • other blood groups have A/B antigens; • immune response if mixed; • causes agglutination of red blood cells;		(5)
(b)	(i)	• (faulty allele) carried/found on female sex/X chromosome; • males affected/female carriers/occurs more in males;		(2)
	(ii)	• blood fails to clot/excessive bleeding;		(1)

Total for Question 9 = 10 marks

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 6 of 7 · Question 5 part b.i · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

5 (a)	(i)	• change in nucleotide/base/codon sequence; • of DNA;		2
	(ii)	• new/different (sequence of) amino acids; • different protein / enzyme / different polypeptide • different phenotype/enzyme does not bind to substrate;		3
(b)	(i)	• allele for haemophilia on X chromosome/ not on Y chromosomes; • hairy ears could only be males / haemophilia could occur in females (and males);		2
	(ii)	• 1 = XX; • 2 = XY^H;		2
	(iii)	• chance of being a girl = 50%/ this can be indicated in a punnet square; • girl will have XX genotype / girl indicated as XX from punnet square/ girls have two X; • no Y chromosome so no chance of hairy ears;		3

Total 12 marks

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 6 of 7 · Question 5 part b.i · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 7 of 7 · Question 6 part a.iii · 5 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

6 (a) (i)	2500 males; $\frac{2500 \times 0.8}{100}$; = 20;		ecf if use 5000 as starting figure 40 scores 2 marks any number×0.8=1	3
(ii)	small number/ not enough tested;			1
(iii)	• many more males than females with condition; • show that it is sex-linked; • to X chromosome; • females are XX /males one X chromosome/XY • females can be carriers(so fewer)/males cannot be carriers(so more); • (relatively) small number (in whole population)/small chance of being colour blind; • indicates that it is recessive;			5
(b)	parents	$X^BY \times X^BX^b$;		4
	gametes	X^B Y X^B X^b;		
	offspring genotypes	$X^B X^B$ X^BX^b X^BY X^bY;		
	offspring phenotypes	normal carrier/ disease female normal female male red-green male;	If parents are incorrect ecf for mp 2,3 and 4 If X and Y are not used- max of 1 mark	
				Total = 13

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 7 of 7 · Question 6 part a.iii · 5 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks